Operating Systems & DBMS Core Technical Interview Capsule

High-yield core computer science revision: Thread vs Process, Deadlocks prevention, Paging, Virtual Memory, B+ Tree indexes, ACID properties, and SQL Transaction Isolation levels.

LITHAVEN CAMPUS STUDY VAULT • OFFICIAL REVISION CAPSULE
COURSE: Core CS Interview Series | **TARGET:** Campus Placement Interviews & GATE Computer Science

1. Operating Systems: Synchronization, Memory & Deadlocks

Concept Area	Key Mechanism & Definition	Underlying Hardware / Kernel Implementation	Interview Question & Answer
Process vs Thread	Process has independent address space (PCB). Threads share text/heap/data segments with own stack and registers.	Context switch between processes flushes TLB; thread context switch preserves TLB.	Q: Why are threads lighter? A: Less state to save; no address space remapping.
Deadlock Conditions	4 Coffman conditions: Mutual Exclusion, Hold and Wait, No Preemption, Circular Wait.	Break Circular Wait by ordering resource acquisition globally (e.g. strict hierarchy).	Q: What is Banker's algorithm? A: Safe state simulation before granting resource requests.
Paging & TLB	Logical address divided into Page Number and Offset. MMU translates page to physical frame.	TLB (Translation Lookaside Buffer) caches recent translations. TLB miss incurs page table walk.	Q: What is thrashing? A: High page-fault frequency where CPU spends more time swapping than computing.
Mutex vs Semaphore	Mutex is binary ownership lock. Semaphore is signaling integer counter (counting/binary).	Wait (P / down) decrements; Signal (V / up) increments counter.	Q: Can thread B unlock mutex owned by A? A: No, mutex requires ownership; semaphore does not.

2. DBMS: Indexing, ACID & SQL Transaction Isolation

Isolation Level	Dirty Read Prevention	Non-Repeatable Read	Phantom Read Prevention	Locking / MVCC Mechanism
Read Uncommitted	NO (Allows dirty reads)	NO	NO	Read without shared lock; highest concurrency, lowest consistency.
Read Committed	YES (Guaranteed)	NO	NO	Shared lock released immediately after statement executes (Postgres default).
Repeatable Read	YES	YES (Guaranteed)	NO (In pure 2PL, yes in InnoDB MVCC)	Shared locks held until transaction completes; MVCC snapshot read.
Serializable	YES	YES	YES (Complete isolation)	Range locks / Predicate locks / Strict 2-Phase Locking (2PL). Lowest concurrency.

3. High-Yield B-Tree vs B+ Tree & Normalization Formulas

- **B+ Tree Advantage over B-Tree:** Leaf nodes are linked in a doubly linked list, enabling fast range scans O(log N + K). Internal nodes only store keys (no records), maximizing branching factor (fan-out).
- **1NF to BCNF Rules:** 1NF (Atomic attributes) | 2NF (1NF + No partial dependency on candidate key) | 3NF (2NF + No transitive dependency) | BCNF (Every determinant X in X->Y must be a superkey).
- **WAL (Write-Ahead Logging):** Changes must be appended to disk log before the dirty buffer page is flushed to the table datafile (ARIES recovery protocol).
- **Clustered vs Non-Clustered Index:** Clustered index physically sorts table rows on disk (only 1 per table, typically Primary Key). Non-clustered index stores leaf pointers back to clustered row ID.